Machine Learning vs. Deep Learning

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Introduction

Machine learning (ML) and deep learning (DL) are two types of computer learning, but they suit different problems. In traditional ML, people select the features the algorithm should consider. Deep learning uses neural networks with many layers that learn the features themselves from raw data. This report examines one example of each from the lesson (Indiana Wesleyan University, n.d.), its real-world application, and why one approach is more suitable than the other.

Example 1: Spam Email Detection (Machine Learning)

The lesson describes a spam filter in which people select features such as particular words, message length, or the sender's domain, and an algorithm uses them to classify each email as spam or not spam. Real email systems use this kind of approach. SpamAssassin, an open-source spam filter, uses a Bayesian classifier that learns from the words found in emails categorized as spam or not spam.

Machine learning is suitable here because the useful signals are easy to name. People already know that suspicious phrases, unusual sender domains, and strange formatting are signs of spam, so they can supply those features directly. The data is not especially large or complex, and models such as naive Bayes or logistic regression are fast and inexpensive to train. They are also explainable, since it is possible to see which features caused a message to be marked as spam. That matters when a legitimate email is blocked and someone wants to know why. As spammers change their language, a filter of this type can also be retrained quickly.

Deep learning is not the best approach for this problem. A deep neural network would need much more data and computing power, and it would be hard to tell why it flagged a specific

email. When the important features are already known, the added complexity offers little benefit and can lead to overfitting, which means memorizing the training data instead of learning general rules.

Example 2: Image Recognition (Deep Learning)

Image recognition is one of the most common deep learning applications. A real-world example is the screening of retinal photographs for diabetic retinopathy, an eye disease associated with diabetes. Convolutional neural networks are trained on large collections of retinal images labeled by experts, and an FDA-authorized system called IDx-DR uses this kind of technology to identify patients who need an eye exam.

Deep learning is suitable because the input is raw pixels, and the signs of disease, such as tiny bleeds and subtle changes in blood vessels, are hard for a person to describe as rules. The neural network learns its own features in layers. Early layers pick up edges and textures, and later layers combine them into shapes such as lesions. This ability to learn from raw data makes deep learning powerful for images, audio, and text, and it works best when there is a large amount of labeled data and enough computing power.

Traditional ML is not suitable here because someone would have to design features by hand for each type of lesion in each type of image. That would be slow, and it would probably miss subtle patterns that a deep network can find. Accuracy would likely be much lower than what deep learning achieves on image tasks. Deep learning also has costs. It needs a large number of labeled images and substantial computing resources, and it is harder to explain than a simple model. In a medical setting, this is why a clinician still reviews the results.

Conclusion

The choice between ML and DL depends on the data and on how well the problem is understood. When the data is structured and people can name the features that matter, as in spam detection, simple ML is faster, cheaper, and easier to explain. When the data is raw and complex, as with images, and the patterns are difficult to describe by hand, deep learning is the better choice. A good rule is to use the simpler approach when it works and deep learning when the problem requires it.

References

Indiana Wesleyan University. (n.d.). *Machine learning vs. deep learning* [Course lesson, Workshop 2.3]. AIML-500 Machine Learning Fundamentals. Brightspace.